

Question	Answer	Marks
3(a)	When two or more waves of the same type overlap, the resultant displacement at any point at any instant is given by the vector sum of the individual displacements that each individual wave would cause at that point at that instant.	B2
3(b)	As the reflector moves, the path difference between the direct wave from the source and reflected wave to the detector varies. When the 2 waves meet in antiphase and interfere destructively, the detected intensity is low. When the two waves meet in phase and interfere constructively, the detected intensity is high. Accept for analysis using path difference together with phase change at reflection.	B1 B1 B1
3(c)(i)	path difference = reflected path length – direct path length $= \sqrt{2}(5) - 5 = n(0.59) \Rightarrow n = 3.5$ Since the path difference is odd multiple of half a wavelength and including the phase change due to the reflection, the waves meet in phase/interfere constructively and the detected intensity is a maximum/high.	B1 B1 B1
3(c)(ii)	To determine maximum order: maximum path difference = $5 = n(0.59) \Rightarrow n = 8.5$ The intensity alternates from maximum to minimum to maximum 5 times	B1 B1 B1